# Bruno Santos

Researcher in Statistics

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## Education

**PhD** **University of Sao Paulo**, Statistics — Sao Paulo, Brazil, Jan 2012 – Dec 2016
- Supervisor: Heleno Bolfarine
- Thesis title: Extensions of Bayesian quantile regression models
- Research visit at Duke University (2013-2014) under Alan Gelfand and Surya Tokdar

**MSc** **University of Sao Paulo**, Statistics — Sao Paulo, Brazil, Jan 2010 – Dec 2012
- Supervisor: Silvia Nagib Elian
- Dissertation title: Quantile regression models

**BSc** **University of Sao Paulo**, Statistics — Sao Paulo, Brazil, Jan 2005 – Dec 2008

## Experience

**CEAUL (Centro de Estatistica e Aplicacoes - Universidade de Lisboa)**, Researcher in Statistics — Lisbon, Portugal, Apr 2024 – Dec 2025, 1 year 9 months

**University of Kent**, Lecturer in Statistics — Canterbury, United Kingdom, Sept 2021 – Dec 2023, 2 years 4 months
- Teaching: Statistics I; Probability and Classical Inference (2021/2022); Advanced Statistical Modelling; Predictive Modelling (2022/2023).

**Federal University of Espirito Santo (UFES)**, Assistant Professor — Vitoria, Brazil, Oct 2019 – July 2021, 1 year 10 months
- Teaching: Statistics I; Statistics II; Probability and Statistics I; Analysis of Categorical Data; Computational Statistics II.

**Georg-August-Universitaet Goettingen**, Postdoctoral Researcher — Goettingen, Germany, Feb 2018 – Sept 2019, 1 year 8 months
- Teaching: Generalized Regression; Spatial Statistics.

**Federal University of Bahia (UFBA)**, Assistant Professor — Salvador, Brazil, Dec 2016 – Jan 2018, 1 year 2 months
- Teaching: Statistical Methods; Bayesian Inference.

## Publications

**A defective cure rate quantile regression model for male breast cancer data** — 2025
Rodrigues, A., Borges, P., Santos, B.
(Journal of Applied Statistics, 52(8), 1485-1512)

**Using population register data and capture-recapture models to estimate over-coverage in Sweden** 2024
Santos, B., Mussino, E., Drefahl, S., Matechou, E.
(Scientific Reports 14, 30551)

**Multiple systems estimation for studying over-coverage and its heterogeneity in population registers** 2024
Mussino, E., Santos, B., Monti, A., Matechou, E., Drefahl, S.
(Quality & Quantity 58, 5033-5056)

**Bias reduction of maximum likelihood estimates for an asymmetric class of power models with applications** 2023
Gomez, Y., Santos, B., Gallardo, D., Venegas, O., Gomez, H.
(REVSTAT - Statistical Journal, 21(4), 491-507)

**Noncrossing structured additive multiple-output Bayesian quantile regression models** 2020
Santos, B., Kneib, T.
(Statistics and Computing)

**Bayesian quantile regression analysis for continuous data with a discrete component at zero** 2018
Santos, B., Bolfarine, H.
(Statistical Modelling)

**Bayesian analysis for zero-or-one inflated proportion data using quantile regression** 2015
Santos, B., Bolfarine, H.
(Journal of Statistical Computation and Simulation)

**Influence measures in quantile regression models** 2015
Santos, B., Elian, S.
(Communications in Statistics - Theory and Methods)

**Association of pollution with quantiles and expectations of the hospitalization rate of elderly people by respiratory diseases in the city of Sao Paulo, Brazil** 2014
Alencar, A., Santos, B.
(Environmetrics)

## Conference Proceedings

**Growth curves for multiple-output response variables via Bayesian quantile regression models** 2020
Santos, B., Rodrigues, A., Kneib, T.
(Proceedings of the 35th International Workshop on Statistical Modelling)

**Structured additive multiple-output noncrossing Bayesian quantile regression models** 2018
Santos, B., Kneib, T.
(Proceedings of the 33rd International Workshop on Statistical Modelling)

**Analysis of Brazil's presidential election via Bayesian spatial quantile regression** 2015
Santos, B., Bolfarine, H.
(Proceedings of the 30th International Workshop on Statistical Modelling)

**A two-part model using quantile regression under a Bayesian perspective** 2013
Santos, B., Bolfarine, H.
(Proceedings of the 28th International Workshop on Statistical Modelling)

**Analysis of residuals in quantile regression: an application to income data in Brazil** 2012
Santos, B., Elian, S.
(Proceedings of the 27th International Workshop on Statistical Modelling)

## Awards

**Collaborations with Developing Countries, Grant Scheme 5** 2022
London Mathematical Society (Grant amount: GBP 1,972)

**2nd place, Best poster award** 2018
23rd Brazilian National Symposium of Probability and Statistics

**Best Student Poster Toyota Corporation Award** 2015
30th International Workshop on Statistical Modelling (USD 500)

**Student Travel Grant** 2013
28th International Workshop on Statistical Modelling (EUR 800)

## Teaching

University of Kent (2021/2022): Statistics I; Probability and Classical Inference

University of Kent (2022/2023): Advanced Statistical Modelling; Predictive Modelling

Federal University of Espirito Santo (2019-2021): Statistics I; Statistics II; Probability and Statistics I; Analysis of Categorical Data; Computational Statistics II

Georg-August-Universitaet Goettingen (2018-2019): Generalized Regression; Spatial Statistics

Federal University of Bahia (2017): Statistical Methods; Bayesian Inference

## Short Courses

3rd Paraibano Meeting of Statistics (2021): Regression models beyond the conditional mean

24th Brazilian National Symposium of Probability and Statistics (postponed, 2020): Visualization of spatial effects in regression models

XIII School of Regression Models (2013): Regression models with truncated and censored responses

Universidad de Antofagasta (2017): Quantile regression: a classical and a Bayesian approach

## Service and Activities

Member, Migrating Trajectories research project (University of Stockholm and University of Kent)

Local organizing committee, Movement and migration in ecology workshop (University of Kent, May 2022)

Local organizing committee, 1st Workshop on Data Science and Statistical Learning (UFES, November 2019)

Leader, Observatory of Covid-19 in favelas of Espirito Santo (OCOFA)

Program Committee member, useR! 2021 and useR! 2022 conferences

## Software

**baquantreg**
R package for Bayesian quantile regression.
- GitHub: https://github.com/brsantos/baquantreg

**ggcrqr**
The goal of ggcrqr is to provide R routines for estimation of Generalized Gompertz Cure Rate Quantile Regression models.
- GitHub: https://github.com/brsantos/ggcrqr

**overcoverage**
Capture-Recapture Hidden Markov Models for register-based population inference.
- Website: https://brsantos.github.io/overcoverage/

**ocmse**
R package used to estimate overcoverage with Multiple System Estimation Models.
- Website: https://brsantos.github.io/ocmse/